

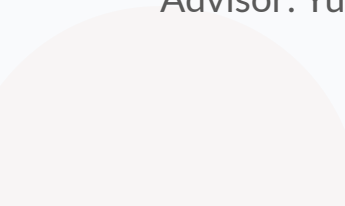


A Traceable Hierarchical Memory System for Long-Term Research Meeting Support

Team 2

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Background

Research meetings are longitudinal and cumulative. Decisions and directions evolve across weeks.

Current AI systems often treat meetings as isolated text blocks, leading to significant challenges in maintaining a coherent research narrative.

Challenges in Long-Term Meeting Support:



Evolving Research Topics

Research ideas and TODOs evolve non-linearly across the research lifecycle.



Outdated vs. Current Directions

Old directions may be updated or abandoned, requiring version control for thoughts.



Source Traceability

Need to trace the origin and rationale behind specific design decisions.

Why Simple Baselines Are Not Enough

Approach	What it does	Limitation
Recent Chat Context	Keeps only recent conversation turns	Forgets earlier decisions and rationale
Flat RAG	Retrieves similar transcript chunks	Misses topic evolution and latest research state
Full Transcript	Places all available transcripts into the prompt	Costly, noisy, context-window limited, and hard to inspect

Comparison of Structured Memory Systems

Type	Examples	Core Idea	Gap for Research Meetings
Fact / Salient Memory	Mem0, MemoryBank	Store salient memories for recall.	Weak in evidence and rationale.
Note-based Memory	A-Mem	Turn interactions into linked notes.	Links may be hard to verify.
Graph-based Memory	Zep / Graphiti, HippoRAG	Model entities and temporal relations.	Relations may miss research decisions.
Hierarchical Memory	H-Mem, MemGPT	Organize memory by hierarchy.	Not focused on evidence tracking.
Evidence-first Memory	Ours	Anchor memory in original evidence.	Targets traceable decisions and evolution.

Key Hypotheses

Core Vision

Long-term research memory should be reusable, inspectable, and correctable.

1

Traceable Evidence

Improves answer grounding and reliability.

2

Topic-State Memory

Improves long-term reasoning.

3

Inspectable Retrieval

Increases human trust and controllability.

How One Query Retrieves Layered Memory

RETRIEVAL TRACE RESULT

What this one query retrieves

The query first finds source-grounded L1 evidence, then uses those seeds to pull the relevant L2 evolution and L3 topic family.

TEASER QUERY

Why did we split transcripts into segments and idea units?

A rationale question about design evolution.

L1 EVIDENCE SEEDS

0422 / 0429 / 0506

0422 L1-0422-006 為了取代逐「輪」(turn)更新記憶體的簡單作法，團隊探討了兩種新的方法來預處理逐字稿以生成「idea units」。第一種是迭代法：系統讀取固定行數的文本，並根據主題連續性逐步合併。第二種是預分割法：利用 LLM 判斷主題...

0429 L1-0429-122 專案放棄了將完整逐字稿一次性提供給模型進行記憶更新的初始方法。原因是上下文長度過大，難以處理所需的 JSON 輸出格式，因此轉而採用分塊處理的策略。

0506 L1-0506-001 團隊確立了目前的文本處理採用由上而下 (top-down) 的架構：「Context Planner」將固定大小的窗口 (如 80 行) 交給「Segment Agent」，後者將其切分為「segments」(較廣泛的相關主題...

L2 TOPIC EVOLUTION

idea-unit generation methods

idea-unit generation methods evolved from 2026-03-25 L1-0325-033: 在文獻回顧中發現，有多篇論文專注於長期會議記錄的摘要，其中一篇較新且以應用為導向的 IEEE 論文特別受到關注。to 2026-05-06 L1-0506-012: 有人批評當前的由上而下 (top-down) 的分段策略 (segment -> idea unit) 不自然，認為由下而上 (bottom-up) 的方法 (idea unit -> theme) 可能更合理。

1 SELECTED EVENT 19 OMITTED

PARENT L3 TOPIC FAMILY

transcript segmentation and idea-unit coverage

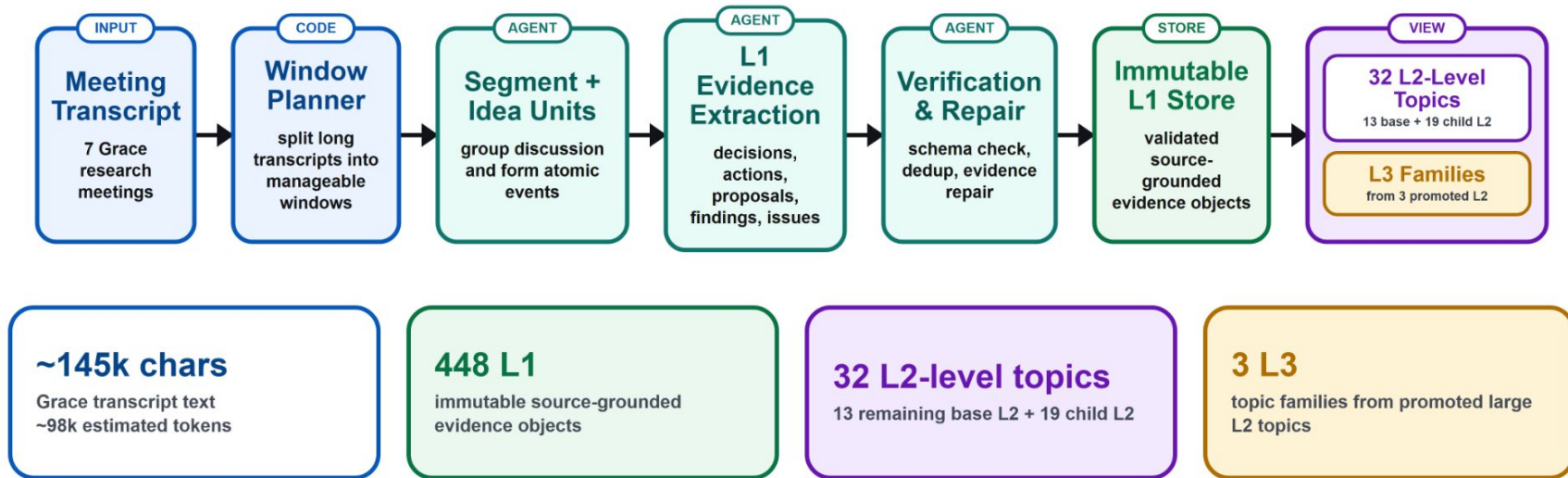
Navigation context: shows this rationale belongs to a broader transcript segmentation / idea-unit coverage family.

FIXED VS DYNAMIC CHUNKING

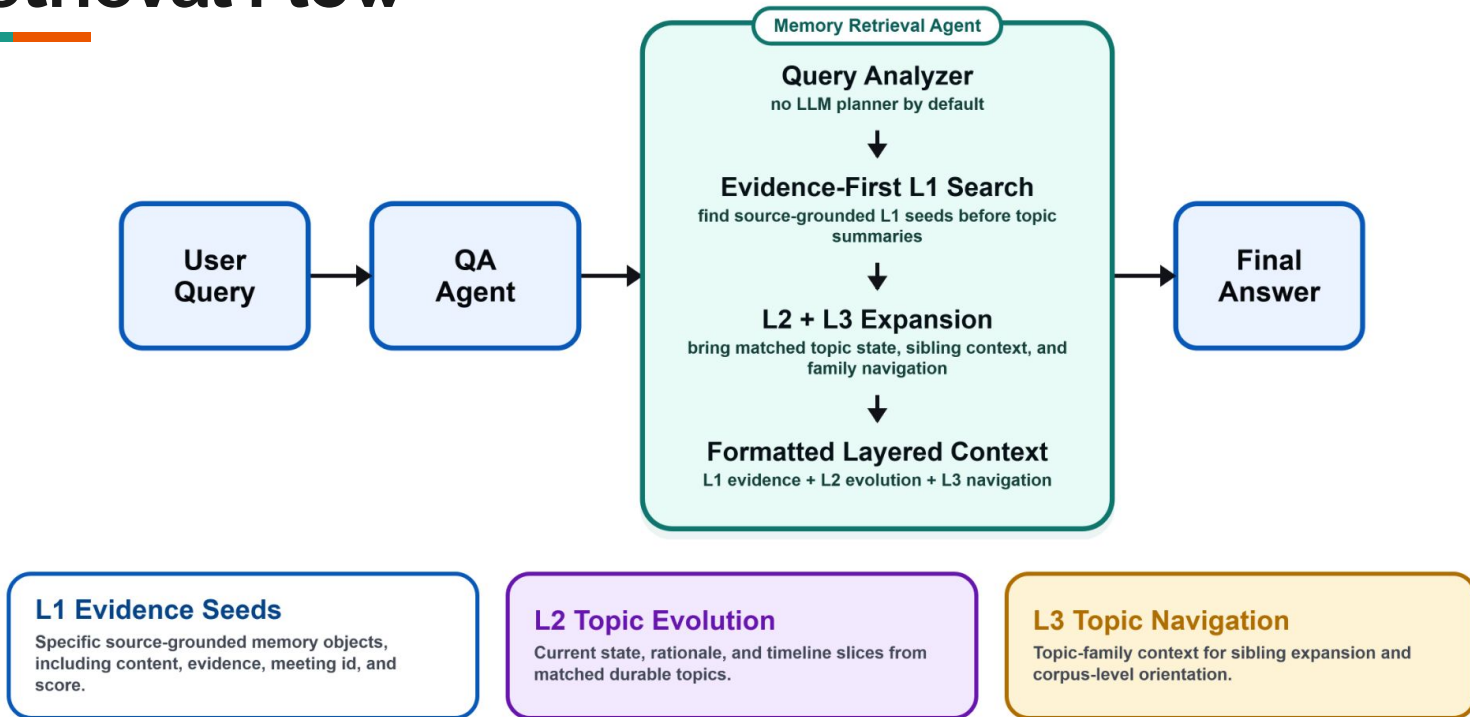
WINDOW AND BOUNDARY SELECTION

TOOL-CALLING TRANSCRIPT READING

Memory Construction



Retrieval Flow



Dataset & Evaluation Questions

Dataset	What it contains	Why we use it
Our Research Meeting Corpus	7 real senior-project meetings(~80K words), 448 L1 objects, 32 L2 level topics, 3 L3 groups(promoted from I2)	Main testbed for long-term memory, decision evolution, and cross-meeting QA
ICSI Meeting Corpus	31 regular group meetings of research groups at ICSI , ~72 hours of audio, transcripts, metadata, speaker info	External test for generalization beyond project-specific data

Evaluation Questions

Our Corpus: 14 audited QA questions

ICSI Corpus: 13 held-out source-constrained QA questions

Evaluation Setup

Goal: Evaluate whether traceable hierarchical memory improves long-term research meeting QA compared with retrieval and memory baselines.

Evaluation Questions Types

- Recent / current active state
- Topic synthesis
- Cross-meeting recall
- Decision reasoning

Compared Methods

Structured Memory: Ours (L1+L2+L3)

Mem0 : Salient inferred memory baseline

H-Mem : Hierarchical note-style memory

Flat RAG: Retrieve transcript chunks directly

A-Mem: Atomic note-based memory

LLM-as-a-Judge Evaluation

Evaluation Framework: Each answer is evaluated with 4 common metrics and 1 question-type-specific metric to ensure quality and specificity.

1. Common Metrics (Base Quality)

Metric	What It Checks
Factuality	Consistent with source meeting facts
Facet Coverage	Covers key factual facets expected
Hallucination	Avoids unsupported decisions/rationale
Grounding	Supported by meeting evidence

2. Question-Type-Specific Metric

Question Type	Metric (Tests)
Evidence Lookup	Context Specificity
Corpus Process	Context Specificity
Temporal Evolution	Explain changes/evolution
Knowledge Update	Final Score/Decisions rationale

Final Quality Score = $(0.7 \times \text{Mean of Base Quality}) + (0.3 \times \text{Type-Specific Metric})$

Dataset 1 - Performance Comparison

Overall Results: Structured Memory Performs Best. Our traceable hierarchical memory achieves the highest overall answer quality and the best acceptable-answer rate among all compared methods.

Method Avg. Quality Score Yes Rate Avg. Context Tokens

Method	Avg. Quality Score	Yes Rate	Avg. Context Tokens
Structured Memory	4.709	92.9%	9,360
Mem0 Inferred	4.624	85.7%	7,717
H-Mem Note	4.420	64.3%	5,690
Flat RAG	4.051	57.1%	10,868
A-Mem	3.426	35.7%	11,526

Category Breakdown: Where Structured Memory Helps

Structured Memory Helps Most for Long-Term Reasoning: Structured Memory is especially effective when the question requires tracking topic evolution, decision rationale, and evidence across meetings.

Question Type	Structured Memory	Best Baseline	Interpretation
Topic synthesis	4.883 / 100%	4.367 / 66.7%	Clear advantage in evolution summaries
Decision reasoning	4.910 / 100%	4.910 / 100%	Matches best; stronger traceability
Recent context	4.767 / 100%	4.940 / 100%	Handled well by short-term memory
Cross-meeting recall	3.760 / 50%	4.525 / 100%	Weakness in timeline coverage

Dataset 2 - Performance Comparison

Overall Results: Structured Memory performs best on ICSI, achieving higher quality and reliability than RAG while using fewer context tokens.

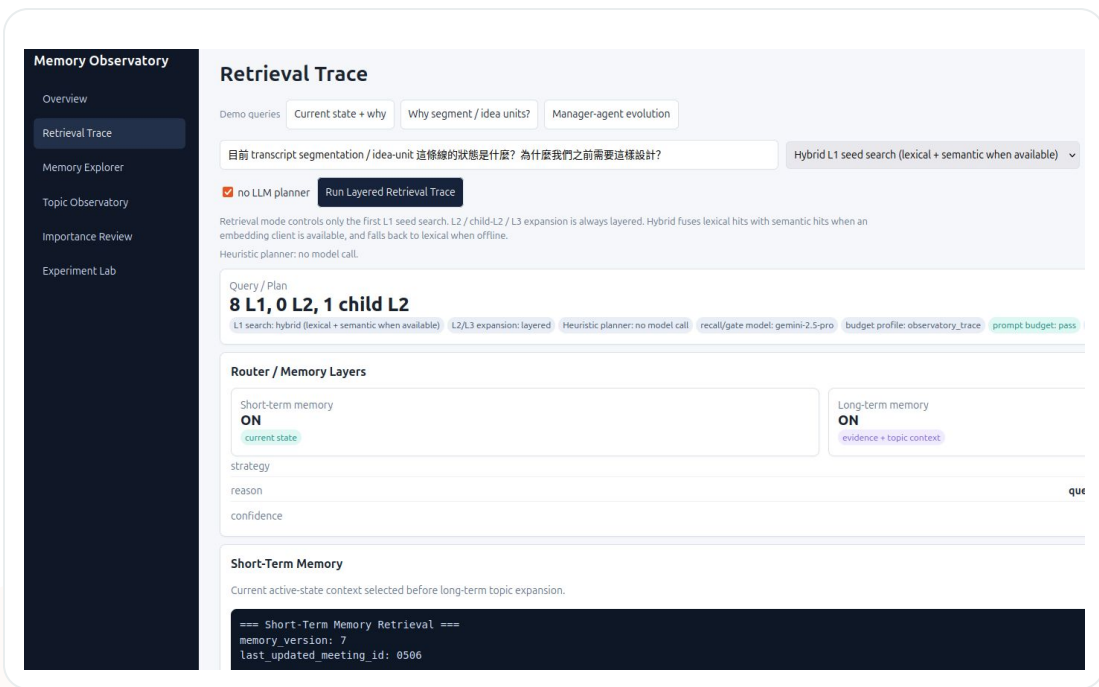
Method Avg. Quality Score Yes Rate Avg. Context Tokens			
Method	Avg. Quality Score	Yes Rate	Avg. Context Tokens
Structured Memory	4.718	69.2%	7.5K
Mem0 Inferred	3.591	0%	3.7K
Flat RAG	3.704	15.4%	11.9K
A-Mem	3.295	0%	9.4K

Category Breakdown: Where Structured Memory Helps

Structured Memory Helps Most for Long-Term Reasoning: Structured Memory outperforms the best baseline on all ICSI question types, with the largest gains in topic synthesis and decision reasoning. Its weakest area is still cross-meeting recall, which suggests future work on improving temporal coverage.

Question Type	Structured Memory	Best Baseline	Interpretation
Topic synthesis	5.000 / 100%	3.795 / 0%	Strongest advantage in topic evolution summaries
Decision reasoning	4.780 / 50%	3.700 / 0%	Higher quality, but acceptable rate drops under stricter judging
Recent context	4.653 / 66.7%	3.997 / 0%	Still better than baselines, but not always fully acceptable
Cross-meeting recall	4.635 / 66.7%	3.988 / 33.3%	Best quality, but timeline/evidence coverage remains weak

Memory Observatory makes retrieval inspectable



The screenshot displays the Memory Observatory web application. On the left is a dark sidebar with navigation links: Overview, Retrieval Trace (selected), Memory Explorer, Topic Observatory, Importance Review, and Experiment Lab. The main content area is titled 'Retrieval Trace' and features several interactive elements. At the top, there are buttons for 'Demo queries', 'Current state + why', 'Why segment / idea units?', and 'Manager-agent evolution'. Below these is a text input field containing a Chinese query about transcript segmentation, followed by a dropdown menu set to 'Hybrid L1 seed search (lexical + semantic when available)'. A checkbox for 'no LLM planner' is checked, and a 'Run Layered Retrieval Trace' button is present. A paragraph explains the retrieval mode: 'Retrieval mode controls only the first L1 seed search. L2 / child-L2 / L3 expansion is always layered. Hybrid fuses lexical hits with semantic hits when an embedding client is available, and falls back to lexical when offline. Heuristic planner: no model call.' Below this, the 'Query / Plan' section shows '8 L1, 0 L2, 1 child L2' and a series of tags: 'L1 search: hybrid (lexical + semantic when available)', 'L2/L3 expansion: layered', 'Heuristic planner: no model call', 'recall/gate model: gemini-2.5-pro', 'budget profile: observatory_trace', and 'prompt budget: pass'. The 'Router / Memory Layers' section contains two boxes: 'Short-term memory' with 'ON' and 'current state', and 'Long-term memory' with 'ON' and 'evidence + topic context'. Below these are fields for 'strategy', 'reason', and 'confidence'. The 'Short-Term Memory' section at the bottom shows the 'Current active-state context selected before long-term topic expansion.' and a code block with the text: '=== Short-Term Memory Retrieval ===', 'memory_version: 7', and 'last_updated_meeting_id: 0506'.

1

Visibility of Logic

Shows query, router decision, L1 seeds, L2 expansion, and context.

2

Sidecar Feedback

Allows inspection and correction without rewriting raw L1 evidence.

3

Auditable Evidence

Turns retrieval from a black box into an auditable chain of evidence.

Memory Observatory

Overview

Retrieval Trace

Memory Explorer

Topic Observatory

Importance Review

Experiment Lab

Retrieval Trace

Demo queries

Current state + why

Why segment / idea units?

Manager-agent evolution

目前 transcript segmentation / idea-unit 這條線的狀態是什麼？為什麼我們之前需要這樣設計？

Hybrid L1 seed search (lexical + semantic when available) ▾

Planner model

heuristic planner

☒ show debug panel

☒ no LLM planner

Run Layered Retrieval Trace

Retrieval mode controls only the first L1 seed search. L2 / child-L2 / L3 expansion is always layered. Hybrid fuses lexical hits with semantic hits when an embedding client is available, and falls back to lexical when offline.

Heuristic planner: no model call.

Query / Plan

8 L1, 0 L2, 1 child L2

L1 search: hybrid (lexical + semantic when available)

L2/L3 expansion: layered

Heuristic planner: no model call

recall/gate model: gemini-2.5-pro

budget profile: observatory_trace

prompt budget: pass

omitted events: 19

Router / Memory Layers

Long-term memory

ON

evidence + topic context

strategy

long_term_only

reason

Memory Observatory is configured to use long-term memory only.

confidence

1

L1 Evidence Seeds

L1-0506-001

0506 | approach_change | score 0.6998

Memory Observatory

Overview

Retrieval Trace

Memory Explorer

Topic Observatory

Importance Review

Experiment Lab

Share Memory Explorer

Meetings

0307

2026-03-07 | 39 objects

6 high importance

0318

2026-03-18 | 25 objects

1 high importance

0325

2026-03-25 | 93 objects

16 high importance

0408

2026-04-08 | 56 objects

4 high importance

0422

2026-04-22 | 47 objects

11 high importance

0429

2026-04-29 | 142 objects

26 high importance

L1 Objects

all types

linked/unlinked

min ir

L1-0307-001

approach_change | imp 0.72

系統採用了一種遺忘機制，如果一個記憶項目在特定次數的近期會議（例如三次）內未被再次討論，該項目將被刪除。

11 linked

L1-0307-002

approach_change | imp 0.71

為簡化實驗，針對實體會議的說話者日誌記錄（speaker diarization）挑戰，將採取假設講者身份的方式進行，除非有像WhisperX這樣高準確度的工具。

11 linked

L1-0307-003

open_issue | imp 0.66

實體會議的說話者分離（speaker diarization）是一項重大挑戰，與會者身份已知的線上會議（如 Zoom）不同。除非有像 WhisperX 這樣高準確度的工具，否則在實驗中可能需要簡化此步驟。

11 linked

L1-0307-004

open_issue | imp 0.61

系統的遺忘機制（例如，如果一個項目在三次會議內未被討論則刪除）

此頁僅在啟動記憶體模型時顯示，防止資訊洩露。

Object Detail

L1-0307-001

approach_change

0307

importance 0.72

系統採用了一種遺忘機制，如果一個記憶項目在特定次數的近期會議（例如三次）內未被再次討論，該項目將被刪除。

Evidence

[SPEAKER_01]: 上一次在 16 這個 meeting 的紀錄都要再往前推，它就會變成一週前、兩週前。可是它現在記錄的是 meeting ID，所以不行。以這個點來說，它的 meeting 現在是 16，更新後就會改成 17；沒有更新時就是 16。因為我們現在設定的是前三次，所以完成 16 之後，如果超過這個範圍就會被處理掉。[SPEAKER_01]: 如果這個點在三次內都沒有被討論，它應該就會把它刪掉。

Topic Links

L3 FAMILY

L3-memory-lifecycle

SOURCE L2

memory lifecycle

ASSIGNED
CHILD L2

importance, activation, and recency

confidence 0.82

related_topic_seed

Technical IDs

Memory Observatory

Overview

Retrieval Trace

Memory Explorer

Topic Observatory

Importance Review

Experiment Lab

Topic Observatory

The tree shows L3 topic families and their child L2 topics. Click an L2 node to inspect its state, timeline, linked L1 evidence, and importance feedback controls.

L2/L3 Topic Tree

filter topics

L3 memory evaluation strategy
L3-memory-evaluation-strategy | 3 child L2 | 34 L1

L2 Overall Memory System Evaluation 12
L2-overall-memory-system-evaluation
Overall Memory System Evaluation: latest state from 2026-04-29 L1-0429-063 is 團隊已確定評估策略，將進行消融研究 (ablation study) 來比較完整系統與多個基準線的效能，包括僅使用 RAG、僅使用短期記憶體，以...

L2 Memory Mechanism Evaluation & Metrics 10
L2-memory-mechanism-evaluation-metrics
Memory Mechanism Evaluation & Metrics: latest state from 2026-04-29 L1-0429-040 is 團隊需要研究並定義 LLM 的「長上下文 (long context)」的具體標準，可能是以 tokens 數量來衡量，以便能適當地評...

L2 Evaluation Datasets & Ground Truth 12
L2-evaluation-datasets-ground-truth
Evaluation Datasets & Ground Truth: latest state from 2026-04-29 L1-0429-043 is 評估方法提議使用一個 LLM 來比對 agent 生成的答案與預設的標準答案 (ground truth)，並以此進行評分。然而，這種使用 ...

L3 memory lifecycle
L3-memory-lifecycle | 4 child L2 | 28 L1

L2 memory retention and forgetting 7
L2-memory-retention-and-forgetting

Overall Memory System Evaluation

L2-overall-memory-system-evaluation

L2 topic child L2 ideal

event_count
12

size_bucket
ideal

parent_l3
**memory
evaluation
strategy**

linked_l1
12

Current State

Overall Memory System Evaluation: latest state from 2026-04-29 L1-0429-063 is 團隊已確定評估策略，將進行消融研究 (ablation study) 來比較完整系統與多個基準線的效能，包括僅使用 RAG、僅使用短期記憶體，以及僅使用長期記憶體。在所有實驗條件下，工作記憶體 (working memory) 將維持不變。Earlier evidence began with 在評估的脈絡下，「Temporal Reasoning」被定義為針對特定會議或對話片段生成摘要或大綱的任務。

Assignment Criteria

evaluation plan ablation baseline system evaluation rag-only short-term-only
long-term-only gold mapping gold mapping table gold event unanswerable
false-premise false premise experiment lab comparison page

Topic Timeline

Memory Observatory

Overview

Retrieval Trace

Memory Explorer

Topic Observatory

Importance Review

Experiment Lab

L2 memory retention and forgetting

7

L2-memory-retention-and-forgetting

memory retention and forgetting: latest state from 2026-04-22 L1-0422-047 is 提出一個選擇性遺忘 (selective forgetting) 的機制，讓大型語言模型 (LLM) 能夠忘記特定敏感資訊 (如真實姓名或電話號碼)，但同時...

Topic Timeline

0318 | L1-0318-008

在評估的脈絡下，「Temporal Reasoning」被定義為針對特定會議或對話片段生成摘要或大綱的任務。

0318 | L1-0318-019

新的「可持續 AI」研究專案將採用新穎的「人性化角度 (human angle)」，探索是否能引導使用者做出更節能的選擇，而非僅專注於硬體或軟體。具體構想包括在聊天機器人中實作「節能模式」。

0325 | L1-0325-011

儘管專案的設定是「導師-學員」(mentor-mentee) 的情境，但使用 ICSI 資料集中的研究會議 (research meetings) 格式是可以接受的，因為「導師-學員」的框架只是一個用於說明的「銷售情境」(selling scenario)，而非對資料來源的嚴格限制。

0325 | L1-0325-016

需要設計一個評估方法，以證明新的人格記憶方法能有效抵抗「人格漂移」(personality drift)。

0325 | L1-0325-027

團隊提出了一種維持 AI 人格特質的方法，即僅使用最近的互動來提取常用語，但如何評估其有效性仍是一個挑戰。目前的評估標準被建議為「真人感」，並提出一個具體的「人在迴路」評估方法：讓原始發言者的學生來評估 agent 的語氣和感覺是否與本人相符。然而，評估方法尚未最終確定。

0325 | L1-0325-069

在評估 AI 分身的真實性時，決議不採用教授本人的自我評估，因其被認為為不可靠的指標。

0325 | L1-0325-084

Contribution



- Proposed a traceable hierarchical memory system for long-term research meetings.
- Built memory with L1 evidence, short-term active state, and L2/L3 topic structure.
- Designed an evidence-first retrieval flow for grounded and inspectable QA.
- Developed Memory Observatory to visualize retrieval traces and support correction.
- Showed better QA quality than RAG and other memory baselines.

Future Work



- Scale evaluation with more meetings, questions, and human validation.
- Improve temporal evolution tracking for latest research direction changes.
- Enhance retrieval ranking to reduce missing-evidence cases.
- Support more interactive human-AI memory inspection and correction.
- Study long-term cost and maintenance as meeting data grows.

Thank You

The slide features a light blue background. In the bottom-left corner, there is a partial view of a light orange circle. In the bottom-right corner, there is a partial view of a light blue circle. The text "Thank You" is centered in a bold, black, sans-serif font.